Data Mining Homework 1

學號：_____　姓名：_____

使用工具：Weka 3.8.6（Apriori）、Python 3（前處理與驗證）

資料前處理

原始資料共 8,002 筆交易、120 項商品，值為 TRUE/FALSE。直接匯入 Weka 會產生大量 FALSE ⇒ FALSE 的規則。

處理方式（preprocess.py）：將 FALSE 改為缺值「?」，每個屬性只保留 {TRUE}，存成 hw1_2026_preprocessed.arff。Weka 不計入缺值，因此只會產生 TRUE ⇒ TRUE 的規則。

Q1

(1)

若 A、B 獨立，$Pr(A \cap B) = Pr(A)Pr(B)$，則 $Conf(A \to B) = Pr(A)Pr(B) / Pr(A) = Pr(B)$。所以只要 B 本身很常被購買，即使 A、B 無關，confidence 仍然很高。

例：1,000 人中 200 人買茶，其中 150 人也買咖啡；全體有 900 人買咖啡。$Conf(Tea \to Coffee) = 0.75$ 看似很高，但 $Pr(Coffee) = 0.9 > 0.75$，買茶反而降低買咖啡的機率（負相關）。只看 confidence 會誤判。

本資料中，beers/ales → soft drinks 的 confidence 為 0.29，而 Pr(soft drinks) = 0.306，lift = 0.95 < 1，同樣是負相關。

(2)

不會。$Lift = Pr(B|A) / Pr(B)$，已除以 Pr(B)，比較的是「有 A 時」與「一般情況下」B 出現的機率。A、B 獨立時 lift 必為 1，不受 Pr(B) 大小影響；上例 lift = 0.83 < 1，可正確看出負相關。

Q2

(1)

$Lift > 1 \Leftrightarrow Pr(B|A) > Pr(B)$：購物籃中有 A 時，B 出現的機率高於平常，A 與 B 正相關。例如 lift = 3 表示買 A 的顧客買 B 的機率是一般顧客的 3 倍，可用於搭配促銷或相鄰陳列。

(2)

- Lift = 1：A、B 獨立，買 A 不影響買 B，搭配銷售沒有額外效果。
- Lift < 1：A、B 負相關，買 A 的人較少買 B，通常是替代品（如兩個品牌的可樂），不適合綁售。

Q3

參數：metric = Confidence，minMetric = 0.9，lowerBoundMinSupport = 0.005，numRules = 10，其餘預設。最終 support = 0.005（40 筆）。

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Apriori
=======

Minimum support: 0 (40 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 20

Generated sets of large itemsets:

Size of set of large itemsets L(1): 120
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Size of set of large itemsets L(2): 2933

Size of set of large itemsets L(3): 8685

Size of set of large itemsets L(4): 7153

Size of set of large itemsets L(5): 2038

Size of set of large itemsets L(6): 155

Best rules found:

 1. crackers/miscellaneous baked food=TRUE frozen pizza=TRUE pasta sauce=TRUE 48 ==> cheese=TRUE 46    <conf:
(0.96)> lift:(5.17) lev:(0) [37] conv:(13.03)
 2. bag snacks=TRUE cheese=TRUE frozen pizza=TRUE hot dogs=TRUE 48 ==> packaged bread/buns/rolls=TRUE 46
<conf:(0.96)> lift:(3.87) lev:(0) [34] conv:(12.04)
 3. cheese=TRUE cold cereal=TRUE cookies/cones=TRUE fluid milk products=TRUE soft drinks=TRUE 46 ==> packaged
bread/buns/rolls=TRUE 44    <conf:(0.96)> lift:(3.87) lev:(0) [32] conv:(11.54)
 4. bag snacks=TRUE cold cereal=TRUE fluid milk products=TRUE salad dressing/sandwich spread=TRUE soft
drinks=TRUE 44 ==> packaged bread/buns/rolls=TRUE 42    <conf:(0.95)> lift:(3.86) lev:(0) [31] conv:(11.04)
 5. bag snacks=TRUE crackers/miscellaneous baked food=TRUE fluid milk products=TRUE pasta sauce=TRUE 43 ==>
cheese=TRUE 41    <conf:(0.95)> lift:(5.14) lev:(0) [33] conv:(11.68)
 6. dry dinner mixes=TRUE eggs=TRUE packaged bread/buns/rolls=TRUE refrigerated juices/drinks=TRUE 42 ==> fluid
milk products=TRUE 40    <conf:(0.95)> lift:(3.52) lev:(0) [28] conv:(10.21)
 7. dry dinner mixes=TRUE eggs=TRUE tropical fruit=TRUE 52 ==> fluid milk products=TRUE 49    <conf:(0.94)>
lift:(3.48) lev:(0) [34] conv:(9.48)
 8. soup=TRUE tropical fruit=TRUE yogurt=TRUE 50 ==> fluid milk products=TRUE 47    <conf:(0.94)> lift:(3.47)
lev:(0) [33] conv:(9.11)
 9. cold cereal=TRUE eggs=TRUE soft drinks=TRUE tropical fruit=TRUE 50 ==> fluid milk products=TRUE 47
<conf:(0.94)> lift:(3.47) lev:(0) [33] conv:(9.11)
10. canned juices=TRUE dry dinner mixes=TRUE eggs=TRUE 49 ==> fluid milk products=TRUE 46    <conf:(0.94)>
lift:(3.47) lev:(0) [32] conv:(8.93)

=== Evaluation ===

Elapsed time: 27.647s
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圖 1　Q3 執行結果

(1) 前 10 條 rules

#	LHS（筆數）	RHS（筆數）	Conf	Lift
1	crackers/miscellaneous baked food frozen pizza pasta sauce (48)	cheese (46)	0.96	5.17
2	bag snacks cheese frozen pizza hot dogs (48)	packaged bread/buns/rolls (46)	0.96	3.87
3	cheese cold cereal cookies/cones fluid milk products soft drinks (46)	packaged bread/buns/rolls (44)	0.96	3.87
4	bag snacks cold cereal fluid milk products salad dressing/sandwich spread soft drinks (44)	packaged bread/buns/rolls (42)	0.95	3.86
5	bag snacks crackers/miscellaneous baked food fluid milk products pasta sauce (43)	cheese (41)	0.95	5.14
6	dry dinner mixes eggs packaged bread/buns/rolls refrigerated juices/drinks (42)	fluid milk products (40)	0.95	3.52
7	dry dinner mixes eggs tropical fruit (52)	fluid milk products (49)	0.94	3.48
8	soup tropical fruit yogurt (50)	fluid milk products (47)	0.94	3.47
9	cold cereal eggs soft drinks tropical fruit (50)	fluid milk products (47)	0.94	3.47
10	canned juices dry dinner mixes eggs (49)	fluid milk products (46)	0.94	3.47

(2) 觀察

後項只有三種熱門商品：fluid milk products（5 條）、packaged bread/buns/rolls（3 條）、cheese（2 條）。前項大致分三類：

- 披薩、義大利麵醬、餅乾 ⇒ 起司（規則 1、5）
- 熱狗、零食、汽水 ⇒ 麵包／熱狗包（規則 2–4）
- 麥片、蛋、水果、果汁、優格 ⇒ 牛奶（規則 6–10）

這些組合在生活中很常見，例如早餐買麥片配牛奶、買熱狗配熱狗麵包、做披薩加起司。不過這些規則的 support 只有 0.5%–0.65%，且後項本身就很熱門，confidence 高有部分是 Q1 所說的原因。

Q4

參數：metric = Confidence，minMetric = 0，只調整 lowerBoundMinSupport。

```
Apriori
=======

Minimum support: 0.12 (952 instances)
Minimum metric <confidence>: 0
Number of cycles performed: 18

Generated sets of large itemsets:

Size of set of large itemsets L(1): 8

Size of set of large itemsets L(2): 1

Best rules found:

 1. packaged bread/buns/rolls=TRUE 1979 ==> fluid milk products=TRUE 957    <conf:(0.48)> lift:(1.79) lev:
(0.05) [421] conv:(1.41)
 2. fluid milk products=TRUE 2167 ==> packaged bread/buns/rolls=TRUE 957    <conf:(0.44)> lift:(1.79) lev:
(0.05) [421] conv:(1.35)

=== Evaluation ===

Elapsed time: 0.53s
```

圖 2　lowerBoundMinSupport = 0.119（952 筆），仍有 2 條規則

```
No large itemsets and rules found!

=== Evaluation ===

Elapsed time: 1.502s
```

圖 3　lowerBoundMinSupport = 0.120（960 筆），找不到規則

(1)

0.120。

(2)

規則至少需要 2 項商品。support 最高的 2-itemset 為 {fluid milk products, packaged bread/buns/rolls}，共 957 筆（0.1196）。門檻 0.120 換算為 960 筆，已高於 957，因此沒有任何 frequent 2-itemset，也就沒有規則。

此值即為所有規則 support 的上限，代表最常一起購買的兩項商品也只出現在約 12% 的交易中，資料相當稀疏。

Q5

130 / 8002 ≈ 0.01625，設 lowerBoundMinSupport = 0.01625（Weka 換算為 130 筆）、minMetric = 0.75、numRules = 1000，在結果中找第一條由 4 項商品組成的規則。

```
Apriori
=======
```

```
Minimum support: 0.02 (130 instances)
Minimum metric <confidence>: 0.75
Number of cycles performed: 20

Generated sets of large itemsets:

Size of set of large itemsets L(1): 120

Size of set of large itemsets L(2): 497

Size of set of large itemsets L(3): 331

Size of set of large itemsets L(4): 39

Size of set of large itemsets L(5): 1

Best rules found:

  1. canned juices=TRUE eggs=TRUE 160 ==> fluid milk products=TRUE 130    <conf:(0.81)> lift:(3) lev:(0.01)
[86] conv:(3.76)
  2. cold cereal=TRUE eggs=TRUE 240 ==> fluid milk products=TRUE 192    <conf:(0.8)> lift:(2.95) lev:(0.02)
[127] conv:(3.57)
  3. bag snacks=TRUE cold cereal=TRUE packaged bread/buns/rolls=TRUE 176 ==> fluid milk products=TRUE 140
<conf:(0.8)> lift:(2.94) lev:(0.01) [92] conv:(3.47)
  4. canned juices=TRUE cold cereal=TRUE 186 ==> fluid milk products=TRUE 147    <conf:(0.79)> lift:(2.92) lev:
(0.01) [96] conv:(3.39)
  5. cheese=TRUE crackers/miscellaneous baked food=TRUE fluid milk products=TRUE 205 ==> packaged
bread/buns/rolls=TRUE 162    <conf:(0.79)> lift:(3.2) lev:(0.01) [111] conv:(3.51)
  6. cold cereal=TRUE packaged bread/buns/rolls=TRUE soft drinks=TRUE 213 ==> fluid milk products=TRUE 168
<conf:(0.79)> lift:(2.91) lev:(0.01) [110] conv:(3.38)
  7. cold cereal=TRUE convenient breakfast/wholesome snacks=TRUE 181 ==> fluid milk products=TRUE 142    <conf:
(0.78)> lift:(2.9) lev:(0.01) [92] conv:(3.3)
  8. cold cereal=TRUE packaged bread/buns/rolls=TRUE tropical fruit=TRUE 194 ==> fluid milk products=TRUE 152
<conf:(0.78)> lift:(2.89) lev:(0.01) [99] conv:(3.29)
  9. bag snacks=TRUE cheese=TRUE fluid milk products=TRUE soft drinks=TRUE 180 ==> packaged
bread/buns/rolls=TRUE 141    <conf:(0.78)> lift:(3.17) lev:(0.01) [96] conv:(3.39)
 10. bag snacks=TRUE hot dogs=TRUE 170 ==> packaged bread/buns/rolls=TRUE 133    <conf:(0.78)> lift:(3.16) lev:
(0.01) [90] conv:(3.37)
 11. cheese=TRUE cold cereal=TRUE packaged bread/buns/rolls=TRUE 244 ==> fluid milk products=TRUE 190    <conf:
(0.78)> lift:(2.88) lev:(0.02) [123] conv:(3.23)
 12. cheese=TRUE hot dogs=TRUE 229 ==> packaged bread/buns/rolls=TRUE 177    <conf:(0.77)> lift:(3.13) lev:
(0.02) [120] conv:(3.25)
 13. bag snacks=TRUE beef=TRUE fluid milk products=TRUE 202 ==> packaged bread/buns/rolls=TRUE 156    <conf:
(0.77)> lift:(3.12) lev:(0.01) [106] conv:(3.23)
 14. beef=TRUE cold cereal=TRUE packaged bread/buns/rolls=TRUE 178 ==> fluid milk products=TRUE 136    <conf:
(0.76)> lift:(2.82) lev:(0.01) [87] conv:(3.02)
 15. cold cereal=TRUE soup=TRUE 182 ==> fluid milk products=TRUE 139    <conf:(0.76)> lift:(2.82) lev:(0.01)
[89] conv:(3.02)
 16. fluid milk products=TRUE hot dogs=TRUE 215 ==> packaged bread/buns/rolls=TRUE 164    <conf:(0.76)> lift:
(3.08) lev:(0.01) [110] conv:(3.11)
 17. cheese=TRUE cold cereal=TRUE soft drinks=TRUE 180 ==> fluid milk products=TRUE 137    <conf:(0.76)> lift:
(2.81) lev:(0.01) [88] conv:(2.98)
 18. eggs=TRUE tropical fruit=TRUE 254 ==> fluid milk products=TRUE 193    <conf:(0.76)> lift:(2.81) lev:(0.02)
[124] conv:(2.99)
 19. cold cereal=TRUE refrigerated juices/drinks=TRUE 245 ==> fluid milk products=TRUE 186    <conf:(0.76)>
lift:(2.8) lev:(0.01) [119] conv:(2.98)
 20. cold cereal=TRUE tropical fruit=TRUE 285 ==> fluid milk products=TRUE 216    <conf:(0.76)> lift:(2.8) lev:
(0.02) [138] conv:(2.97)
 21. cheese=TRUE crackers/miscellaneous baked food=TRUE packaged bread/buns/rolls=TRUE 216 ==> fluid milk
products=TRUE 162    <conf:(0.75)> lift:(2.77) lev:(0.01) [103] conv:(2.86)
 22. baked sweet goods=TRUE fluid milk products=TRUE soft drinks=TRUE 188 ==> packaged bread/buns/rolls=TRUE
141    <conf:(0.75)> lift:(3.03) lev:(0.01) [94] conv:(2.95)
 23. deli counter meats=TRUE tropical fruit=TRUE 184 ==> packaged bread/buns/rolls=TRUE 138    <conf:(0.75)>
lift:(3.03) lev:(0.01) [92] conv:(2.95)
```

```
=== Evaluation ===

Elapsed time: 2.564s
```

圖 4　Q5 執行結果（規則 1、2 只有 3 項商品，規則 3 為第一條 4-itemset 規則）

答：{bag snacks, cold cereal, packaged bread/buns/rolls} ⇒ {fluid milk products}，confidence = 140/176 = 0.7955，lift = 2.94。

另以 q5_check.py 列出 39 個 4-itemsets 的全部 546 條規則驗證，結果相同。

Q6

參數：metric = Lift，minMetric = 3.0，lowerBoundMinSupport = 0.01，numRules = 10。support 下界比 Q3 稍高，以避免極少見的組合排在前面。

```
Apriori
=======

Minimum support: 0.01 (80 instances)
Minimum metric <lift>: 3
Number of cycles performed: 20

Generated sets of large itemsets:

Size of set of large itemsets L(1): 120

Size of set of large itemsets L(2): 1229

Size of set of large itemsets L(3): 1469

Size of set of large itemsets L(4): 485

Size of set of large itemsets L(5): 41

Best rules found:

 1. packaged bread/buns/rolls=TRUE pasta sauce=TRUE 229 ==> beef=TRUE dry noodles/pasta=TRUE 82    conf:(0.36)
< lift:(16.56)> lev:(0.01) [77] conv:(1.51)
 2. beef=TRUE dry noodles/pasta=TRUE 173 ==> packaged bread/buns/rolls=TRUE pasta sauce=TRUE 82    conf:(0.47)
< lift:(16.56)> lev:(0.01) [77] conv:(1.83)
 3. cheese=TRUE pasta sauce=TRUE 228 ==> beef=TRUE dry noodles/pasta=TRUE 81    conf:(0.36) < lift:(16.43)>
lev:(0.01) [76] conv:(1.51)
 4. beef=TRUE dry noodles/pasta=TRUE 173 ==> cheese=TRUE pasta sauce=TRUE 81    conf:(0.47) < lift:(16.43)>
lev:(0.01) [76] conv:(1.81)
 5. beef=TRUE pasta sauce=TRUE 194 ==> cheese=TRUE dry noodles/pasta=TRUE 81    conf:(0.42) < lift:(15.12)>
lev:(0.01) [75] conv:(1.65)
 6. cheese=TRUE dry noodles/pasta=TRUE 221 ==> beef=TRUE pasta sauce=TRUE 81    conf:(0.37) < lift:(15.12)>
lev:(0.01) [75] conv:(1.53)
 7. pasta sauce=TRUE 387 ==> beef=TRUE dry noodles/pasta=TRUE packaged bread/buns/rolls=TRUE 82    conf:(0.21)
< lift:(14.62)> lev:(0.01) [76] conv:(1.25)
 8. beef=TRUE dry noodles/pasta=TRUE packaged bread/buns/rolls=TRUE 116 ==> pasta sauce=TRUE 82    conf:(0.71)
< lift:(14.62)> lev:(0.01) [76] conv:(3.15)
 9. beef=TRUE pasta sauce=TRUE 194 ==> dry noodles/pasta=TRUE packaged bread/buns/rolls=TRUE 82    conf:(0.42)
< lift:(14.45)> lev:(0.01) [76] conv:(1.67)
10. dry noodles/pasta=TRUE packaged bread/buns/rolls=TRUE 234 ==> beef=TRUE pasta sauce=TRUE 82    conf:(0.35)
< lift:(14.45)> lev:(0.01) [76] conv:(1.49)

=== Evaluation ===

Elapsed time: 12.4s
```

圖 5　Q6 執行結果

前 10 條 rules

#	LHS（筆數）	RHS（筆數）	Conf	Lift
1	packaged bread/buns/rolls pasta sauce (229)	beef dry noodles/pasta (82)	0.36	16.56
2	beef dry noodles/pasta (173)	packaged bread/buns/rolls pasta sauce (82)	0.47	16.56
3	cheese pasta sauce (228)	beef dry noodles/pasta (81)	0.36	16.43
4	beef dry noodles/pasta (173)	cheese pasta sauce (81)	0.47	16.43
5	beef pasta sauce (194)	cheese dry noodles/pasta (81)	0.42	15.12
6	cheese dry noodles/pasta (221)	beef pasta sauce (81)	0.37	15.12
7	pasta sauce (387)	beef dry noodles/pasta packaged bread/buns/rolls (82)	0.21	14.62
8	beef dry noodles/pasta packaged bread/buns/rolls (116)	pasta sauce (82)	0.71	14.62
9	beef pasta sauce (194)	dry noodles/pasta packaged bread/buns/rolls (82)	0.42	14.45
10	dry noodles/pasta packaged bread/buns/rolls (234)	beef pasta sauce (82)	0.35	14.45

與 Q3 的差異

	Q3（Confidence）	Q6（Lift）
商品	早餐、熱狗、披薩等多種組合	集中在牛肉＋義大利麵＋義大利麵醬（＋起司、麵包）
後項	單一熱門商品	2–3 項、較不熱門的商品
Conf	0.94–0.96	0.21–0.71
Lift	3.47–5.17	14.45–16.56
形式	各條不同	A⇒B、B⇒A 成對出現（lift 對稱）

Confidence 偏向後項為熱門商品的規則；lift 排除了熱門程度的影響，找出的是互補性強的組合（同一道菜的食材），confidence 雖低，但更適合用於搭配促銷。

Q7

- 工具：Weka 3.8.6 的 Apriori；Python 3 用於前處理（preprocess.py）與 Q5 驗證（q5_check.py）。
- 前處理：FALSE 改為缺值，屬性設為 {TRUE}，輸出 .arff 檔。
- 參數：如下表，其餘（delta = 0.05、upperBoundMinSupport = 1.0）為預設值。

題目	metric	minMetric	lowerBoundMinSupport	numRules
Q3	Confidence	0.9	0.005	10
Q4	Confidence	0	0.119 / 0.120	10
Q5	Confidence	0.75	0.01625	1000
Q6	Lift	3.0	0.01	10

- Q4 先用 Python 算出最大 2-itemset support（957 筆），再於 Weka 驗證 0.119 與 0.120。